An Executable Verification Artifact for the Sawin et al. Lower Bound on the Erdős Unit-Distance Exponent

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Abstract

We present `erdos-ant-verification`, an open-source Python implementation that computationally reproduces the finite, explicitly checkable parts of the 2026 OpenAI/Sawin disproof of the Erdős planar unit-distance conjecture [Ope26; Alo+26]. The central result is a numerical match to equation (2.2) of the remarks paper [Alo+26]: for the explicit construction with $T = \{3, 5, 7, 11, 13, 17\}$, $S = \{101, \infty\}$, and $L_T = \mathbb{Q}(\sqrt{5}, \sqrt{13}, \sqrt{17}, \sqrt{21}, \sqrt{33})$, we compute the exponent excess to be 6.2391×10^{-38}, agreeing with the published value $\approx 6.24 \times 10^{-38}$ to within 0.014% relative error.

This artifact does *not* contain a new mathematical proof, does *not* construct the infinite Golod–Shafarevich tower required for the asymptotic result, and has *not* been peer-reviewed. It is intended as a citable, CI-verified reproducibility artifact for the published numerical lower bound, not as an independent contribution to discrete geometry. The artifact is available at `https://github.com/Flamehaven-Labs/erdos-ant-verification` under the MIT licence.

Reproduction. `pip install erdos-ant-verification && python -m erdos_ant.verify` runs the unit tests and prints the verdict; the same command with `--write-evidence` also refreshes the tracked JSON certificate at `reports/verification_result.json`. Pinned to commit <COMMIT_HASH>[1] and Zenodo <ZENODO_DOI>[2].

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*Project lead, Flamehaven Initiative. Code distribution: github.com/Flamehaven-Labs.

[1] Filled in at release-tag time; see GitHub release `v0.1.3`.

[2] For this artifact's first paper-bearing release `v0.1.3`. The predecessor source-only release `v0.1.1` carries DOI 10.5281/zenodo.20366884. The intermediate release `v0.1.2` was a pre-PDF snapshot held open by external audit.

Contents

1 Background and scope

1.1 The Erdős unit-distance problem

For a finite set $P \subset \mathbb{R}^2$, let $\nu(P) = \#\{\{x, y\} \subset P : |x - y| = 1\}$ and $\nu(n) = \max_{|P|=n} \nu(P)$. The planar unit-distance problem [Erd46] concerns the asymptotic behaviour of $\nu(n)$. Erdős proved $\nu(n) \geq n^{1+\Omega(1/\log\log n)}$ via a square-grid construction and conjectured the matching upper bound $\nu(n) \leq n^{1+C/\log\log n}$ for some absolute constant C.

1.2 The 2026 disproof

The conjecture was disproved in May 2026 by an internal reasoning model at OpenAI [Ope26], with a simplified construction and explicit numerical lower bound published the following day by Alon, Bloom, Gowers, Litt, Sawin, Shankar, Tsimerman, Wang, and Matchett Wood [Alo+26]. The combined result is:

Theorem 1.1 of [Alo+26]. *There exists an absolute constant $\delta > 0$ and infinitely many positive integers n such that $\nu(n) \geq n^{1+\delta}$.*

The construction passes through a CM field $K_j = L_j(i)$, where the L_j form an infinite tower of totally real fields of bounded root discriminant, supplied by the Golod–Shafarevich theorem [GS64]. Section 2 of [Alo+26] gives an explicit choice with $T = \{3, 5, 7, 11, 13, 17\}$, $S = \{101, \infty\}$, and $L_T = \mathbb{Q}(\sqrt{5}, \sqrt{13}, \sqrt{17}, \sqrt{21}, \sqrt{33})$, and computes (eq. (2.2)) a rigorous numerical lower bound $\delta \gtrapprox 6.24 \times 10^{-38}$.

1.3 What this artifact verifies

This paper accompanies a Python package, `erdos-ant-verification`, whose purpose is narrow and explicitly bounded:

- we *do not* re-prove Theorem 1.1;
- we *do not* construct the infinite Golod–Shafarevich tower;
- we *do* re-implement the finite, explicitly checkable parts of the construction in pure Python, with exact small-integer arithmetic for the algebraic-number-theory layer and `mpmath` at 200-bit precision for the numerical evaluation of eq. (2.2);
- we *do* pin the published numerical value $\approx 6.24 \times 10^{-38}$ as a regression test that fails if our code drifts from the published source.

The artifact is therefore neither a proof checker (it does not formalise the proof in a trusted kernel) nor an independent mathematical contribution (it produces no new theorem). It is a reproducibility artifact in the sense of, e.g., the ACM Artifact Evaluation guidelines: a citable, version-pinned, CI-verified executable that allows any third party to re-derive the published numerical value on their own hardware without relying on our reading of the source PDFs.

The remainder of the paper describes the three implementation phases (§§2–4), the central numerical reproduction (§5), the explicit limits (§6), and the reproducibility surface (§7).

2 Phase 1: faithful $h(K) = 1$ lattices

Phase 1 implements the trivial-class-number degenerate case of Lemma 2.2 of [Ope26]: an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$ with class number $h(K) = 1$, where the ideal-class pigeonhole reduces to the identity. For these fields the ring of integers $\mathcal{O}_K$ embed standardly in $\mathbb{R}^2$ as a rank-2 lattice, and unit distances between lattice points can be counted exactly.

2.1 Coverage

We implement the three smallest $h = 1$ cases (the Baker–Heegner–Stark list $1, 2, 3, 7, 11, 19, 43, 67, 163$ is encoded in the module-level constant `CLASS_NUMBER_ONE_DISCRIMINANTS`; the first three are implemented):

d	$\mathcal{O}_K$	Standard embedding into $\mathbb{R}^2$
1	$\mathbb{Z}[i]$ (Gaussian integers)	$a + bi \mapsto (a, b)$
2	$\mathbb{Z}[\sqrt{-2}]$	$a + b\sqrt{-2} \mapsto (a, b\sqrt{2})$
3	$\mathbb{Z}[\omega]$, $\omega = \frac{-1+\sqrt{-3}}{2}$	$a + b\omega \mapsto (a - b/2,\ b\sqrt{3}/2)$

For each d and bound $b \geq 0$, the module enumerates all $(2b + 1)^2$ lattice points with integer coefficients in $[-b, b]$ and counts unit-distance pairs exactly via a numpy $O(n^2)$ broadcast.

2.2 Verifications

Verification 2.1. *For $\mathbb{Z}[i]$ with $b = 1$ (the 3×3 Gaussian grid, $n = 9$), the artifact computes exactly 12 unit-distance pairs: three rows of two horizontal edges plus three columns of two vertical edges. Pinned in `tests/test_imaginary_quadratic_lattice.py::test_gaussian_3x3_grid_has_12_unit_distances`.*

Verification 2.2. *For $\mathbb{Z}[i]$ with $b = 3$ (the 7×7 Gaussian grid, $n = 49$), the artifact computes exactly $2 \cdot 7 \cdot 6 = 84$ unit-distance pairs. Source: `src/erdos_ant/imaginary_quadratic_lattice.py`, class `ImaginaryQuadraticLattice.count_unit_distance_pairs`. Pinned in `src/erdos_ant/verify.py` as the assertion `phase1_gaussian_7x7_grid_has_expected_84_pairs`.*

Verification 2.3. *For each $b \in \{1, 2, 3, 4\}$, the Eisenstein-lattice (triangular) unit-distance count strictly exceeds the Gaussian-lattice count at the same b. The triangular lattice has 6 nearest neighbours per interior point versus the square lattice's 4. Pinned in `test_eisenstein_unit_distances_exceed_gaussian_for_same_bound`.*

2.3 Non-claims

Phase 1 does not improve the asymptotic Erdős exponent. With $[K : \mathbb{Q}] = 2$, the construction is the classical square or triangular grid; in particular $\nu(n) \leq n^{1+O(1/\log\log n)}$ holds and the $\delta > 0$ excess is not present. The asymptotic result strictly requires $[K : \mathbb{Q}] \to \infty$, which is the Phase 3 setting.

Phase 1 is included in the artifact for two reasons: (i) it is a self-contained, exact reference against which the Phase 3 construction can be sanity-checked at low parameter values, and (ii) it documents the standard complex embedding of $\mathcal{O}_K$ in $\mathbb{R}^2$ unambiguously, which is the substrate of every later step.

3 Phase 2: $K = \mathbb{Q}(\sqrt{-5})$ with genus theory

Phase 2 implements the smallest non-trivial-class-number case $K = \mathbb{Q}(\sqrt{-5})$, $h(K) = 2$. The Lemma 2.2 pigeonhole of [Ope26] becomes non-vacuous here: when an ideal product $A_\varepsilon = \prod_j P_j^{\varepsilon_j} (P'_j)^{k-\varepsilon_j}$ lands in the non-principal class, no element $\alpha_\varepsilon \in K^*$ with $(\alpha_\varepsilon) = A_\varepsilon A_\eta^{-1}$ exists for arbitrary η — the pigeonhole must restrict to a single ideal class, and the bound $|U| \geq (k + 1)^s/h(K)$ is sharp.

3.1 Classical facts encoded

- $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ with discriminant -20 (`K_DISCRIMINANT = -20`, `CLASS_NUMBER = 2`).
- The non-trivial ideal class is represented by $\mathfrak{p}_2 = (2, 1+\sqrt{-5})$, satisfying $\mathfrak{p}_2^2 = (2)$.
- By genus theory [Cox89, Theorem 6.1], the Hilbert class field is the genus field $H = K(\sqrt{-1}) = \mathbb{Q}(i, \sqrt{5})$.
- A rational prime p (odd, $p \neq 5$) splits completely in K iff $\left(\frac{-5}{p}\right) = 1$, equivalently $p \bmod 20 \in \{1, 3, 7, 9\}$. Of these, $p \bmod 20 \in \{1, 9\}$ gives a principal prime (representable as $a^2 + 5b^2$), and $p \bmod 20 \in \{3, 7\}$ gives a non-principal prime.

3.2 Lemma 2.2 pigeonhole, explicit

The Phase 2 module `erdos_ant.genus_class_field` exposes a `GenusClassFieldProbe` that, given a tuple of split rational primes, computes the ideal-class partition of the $\{0,1\}^s$ family of ideal products and reports $|U| \geq (k+1)^s / h(K)$.

Verification 3.1. *For the choice* $(p_1, p_2) = (29, 41)$ *(both in the principal genus,* $29 = 3^2 + 5 \cdot 2^2$, $41 = 6^2 + 5 \cdot 1^2$*), all four candidate ideals* A_ε *for* $\varepsilon \in \{0,1\}^2$ *are principal, and the Lemma 2.2 lower bound is* $(1+1)^2/2 = 2$*. Pinned in* `tests/test_genus_class_field.py::test_lemma_22_pigeonhole_with_two_principal_primes`*.*

Verification 3.2. *For the choice* $(p_1, p_2) = (29, 3)$ *(one principal, one non-principal), the four ideals partition* $2 + 2$ *between the two ideal classes, and the Lemma 2.2 lower bound is still* 2*. Pinned in* `test_lemma_22_pigeonhole_with_mixed_genus`*.*

3.3 Non-claims

Phase 2 fixes a single CM field of degree 2 over $\mathbb{Q}$ and produces a finite pigeonhole demonstration. It does not produce a sequence of fields with $[K_j : \mathbb{Q}] \to \infty$, and does not generate the exponentially growing set U that drives the asymptotic exponent excess. It does, however, instantiate the smallest concrete example where the pigeonhole step is genuinely non-trivial.

4 Phase 3: the Sawin multi-quadratic construction, finite parts

Phase 3 implements the finite, explicitly checkable parts of the *simplified* construction of Section 2.1 of [Alo+26], which uses an unramified pro-2 tower over a multi-quadratic base field rather than the original GPT pro-3 tower over a cyclic cubic. The simplification is due to Sawin and yields the explicit numerical bound evaluated in §5.

4.1 The Sawin parameters, verbatim

The construction fixes

$$\begin{aligned} T &= \{3,\ 5,\ 7,\ 11,\ 13,\ 17\}, \\ S &= \{101,\ \infty\}, \\ L_T &= \mathbb{Q}(\sqrt{5},\ \sqrt{13},\ \sqrt{17},\ \sqrt{21},\ \sqrt{33}), \end{aligned}$$

with $[L_T : \mathbb{Q}] = 2^5 = 32$, and sets $K = L_T(i)$ as the CM field.

These values appear verbatim in the artifact:

- `T_SET = (3, 5, 7, 11, 13, 17)`

- `S_SET_SPLIT = (101,)` (the infinite place is implicit)
- `L_T_GENERATOR_RADICANDS = (5, 13, 17, 21, 33)`
- `SAWIN_R_PRODUCT = 2 · 3 · 5 · 7 · 11 · 13 · 17 = 510510`

and the test suite pins each constant against the published source.

4.2 Splitting of 101 in L_T

A rational prime p splits completely in $\mathbb{Q}(\sqrt{d_1}) \cdot \mathbb{Q}(\sqrt{d_2}) \cdots \mathbb{Q}(\sqrt{d_s})$ if and only if it splits completely in each factor $\mathbb{Q}(\sqrt{d_i})$ (which, for linearly disjoint quadratic extensions, is the definition of completely splitting in the compositum). For odd $p \neq d_i$, splitting in $\mathbb{Q}(\sqrt{d_i})$ is equivalent to $\left(\frac{d_i}{p}\right) = 1$.

Verification 4.1. *The artifact evaluates the Jacobi symbol* $\left(\frac{d_i}{101}\right)$ *for each* $d_i \in \{5, 13, 17, 21, 33\}$ *and finds all five symbols equal to* $+1$*. Therefore* 101 *splits completely in* L_T*. Source:* `src/erdos_ant/sawin_multiquadratic.py`*, function* `splits_completely_in_L_T`*. Pinned in* `tests/test_sawin_multiquadratic.py::test_101_splits_completely_in_each_Q_sqrt_d`*.*

4.3 Galois-rank counts and Golod–Shafarevich admissibility

Let $G = \mathrm{Gal}(\widetilde{\mathbb{Q}}_T/\mathbb{Q})$ be the Galois group of the maximal pro-2 extension of $\mathbb{Q}$ unramified outside T. Because T contains a prime that is $3 \pmod 4$ (e.g. 3, 7, or 11), the 2-rank of the maximal elementary abelian unramified quotient is $|T| - 1 = 5$, giving

$$d(G_T^\infty) = |T| - 1 = 5.$$

Adjoining S adds at most $|S| - 1 = 1$ Frobenius relations (Sawin's pro-2 analogue of the Hajir–Maire–Ramakrishna cut), so

$$r(G_T^S) \le d(G_T^\infty) + |S| - 1 = 5 + 1 = 6.$$

The Golod–Shafarevich admissibility condition for the tower to remain infinite is $r \le d^2/4$; here

$$r(G_T^S) \le 6 \le \frac{(|T| - 1)^2}{4} = \frac{25}{4} = 6.25,$$

which holds strictly. By the cited Golod–Shafarevich theorem [GS64], this admissibility implies the tower is infinite. The artifact verifies the admissibility inequality only; it does *not* itself prove or compute the infinitude — the cited theorem does.

Verification 4.2. *The artifact computes* $d(G_T^\infty) = 5$*,* $r(G_T^S) \le 6$*, and the Golod–Shafarevich threshold* $25/4 = 6.25$*, and confirms* $6 \le 6.25$*. Pinned in* `test_d_G_infty_equals_T_minus_1_because_T_contains_3_mod_4`*,* `test_r_bound_is_six_per_remarks_pdf`*, and* `test_golod_shafarevich_inequality_holds`*.*

4.4 Non-claims

Phase 3 does *not* construct the infinite tower itself. The existence of the tower is a non-constructive consequence of the Golod–Shafarevich theorem [GS64] together with Sawin's pro-2 analogue of the Hajir–Maire cut [HM01; Alo+26]; no finite computation can exhibit a closed-form description of L_j for arbitrary j. Phase 3 checks every step of the construction that *is* amenable to finite computation, and stops at that boundary.

5 Numerical reproduction of equation (2.2)

This is the central numerical content of the artifact. We reproduce equation (2.2) of [Alo+26] as a regression test against the published source.

5.1 The published expression

Sawin et al. state immediately after Lemma 2.1 [Alo+26, §2.1, eq. (2.2)]:

> *Explicitly, the exponent in (2.1) can be taken to be*
>
> $$1 + \frac{\log(u\pi/(36v))}{\log(36/\delta^2)} \approx 1 + 6.24 \cdot 10^{-38}$$
>
> *where $v = r/2$, $\delta \geq 101^{-2\lceil 18r^3/\pi \rceil}$, and $u = \lceil 18r^3/\pi \rceil \cdot r^{-2}$, with $r = 2{\cdot}3{\cdot}5{\cdot}7{\cdot}11{\cdot}13{\cdot}17$.*

Let $K = \lceil 18r^3/\pi \rceil$, so that $u = K/r^2$ and $\delta = 101^{-2K}$. Direct substitution gives

$$\frac{u\pi}{36v} = \frac{(K/r^2) \cdot \pi}{36 \cdot (r/2)} = \frac{K\pi}{18r^3},$$

and the exponent excess above 1 simplifies to

$$\delta_{\text{excess}} = \frac{\log(K\pi/(18r^3))}{\log 36 + 4K \log 101}. \tag{1}$$

5.2 Why `float64` fails

The numerator of (1) is $\log(K\pi/(18r^3))$. By definition of the ceiling, $K \geq 18r^3/\pi$, so the ratio $K\pi/(18r^3)$ lies in the half-open interval

$$\left(1,\ 1 + \frac{\pi}{18r^3}\right].$$

For $r = 510510$, $r^3 \approx 1.33 \cdot 10^{17}$, hence $\pi/(18r^3) \approx 1.31 \cdot 10^{-18}$.

The numerator is therefore $\log(1 + \varepsilon)$ for some $\varepsilon \in (0, 1.31 \cdot 10^{-18}]$. IEEE-754 `float64` has a 52-bit mantissa and machine epsilon $\approx 2.22 \cdot 10^{-16}$, so it cannot distinguish $1 + 1.3 \cdot 10^{-18}$ from 1 in the first place. The naive evaluation rounds the ratio to exactly 1, yields $\log 1 = 0$, and reports $\delta_{\text{excess}} = 0$.

This is not a flaw in eq. (1); it is a property of finite-precision arithmetic. Any honest evaluation must use a precision significantly above 1 part in 10^{18}.

5.3 Evaluation in `mpmath` at 200-bit precision

The artifact uses `mpmath` [The23] with `mp.prec = 200` (about 60 decimal digits) to evaluate the quantities in eq. (1) exactly enough that the $1 + 10^{-18}$ separation survives. The function `evaluate_sawin_exponent_bound` in `erdos_ant.sawin_multiquadratic` returns a `SawinDeltaEvaluation` dataclass containing r, K, v, $\log u$, and δ_{excess}.

Verification 5.1. *Calling* `evaluate_sawin_exponent_bound()` *(source:* `src/erdos_ant/sawin_multiquadratic.py`*, function* `evaluate_sawin_exponent_bound`*) returns*

$$r = 510{,}510, \qquad K = \lceil 18 \cdot 510510^3/\pi \rceil \approx 7.62 \cdot 10^{17},$$

and

$$\delta_{excess} = 6.2391 \cdot 10^{-38}.$$

The published value is $\approx 6.24 \cdot 10^{-38}$ *(given to two significant figures in [Alo+26]). The relative error between the artifact's evaluation and the published value is*

$$\frac{|6.2391 - 6.24|}{6.24} = 1.4 \cdot 10^{-4} = 0.014\%,$$

fully consistent with two-significant-figure rounding in the source (see also `docs/DEVIATION_LOG.md` *for the full deviation ledger). Pinned in* `tests/test_sawin_multiquadratic.py::test_sawin_exponent_bound_matches_remarks_eq_2_2`*. The per-source-file SHA-256 manifest under which this number was computed appears in* `reports/verification_result.json` *field* `source_sha256_manifest`*.*

5.4 What the verification establishes (and does not)

Verification 5.1 establishes that, given the inputs T, S, L_T, K exactly as written in [Alo+26, §2.1], the elementary closed-form expression eq. (1) evaluates to the value the source paper claims, to within published rounding.

This does *not* establish:

- that the closed-form expression is itself correct (that depends on Lemma 2.1 and earlier algebra of [Alo+26], which we do not re-derive);
- that the underlying infinite Golod–Shafarevich tower exists (Phase 3 only verifies the finite admissibility condition);
- that $\delta_{\text{excess}} = 6.24 \cdot 10^{-38}$ is in any sense optimal — it is one explicit lower bound, not the best possible.

What it does establish is that any future code change in the artifact that drifts away from the published value will fail the regression test at the 0.5 relative-tolerance band hard-coded in the test (currently 0.014%, well below the 50% tolerance the test uses).

6 Limits and explicit non-claims

This section is placed before reproducibility deliberately. The shortest honest summary of this artifact is the list of things it does not do.

6.1 This artifact does NOT contain a new proof

Theorem 1.1 of [Ope26; Alo+26] is proved by those authors. The present paper does not re-derive Lemma 2.1, Lemma 2.2, Proposition 2.3, or any other intermediate result. Where the source papers say "by the Golod–Shafarevich theorem" or "by the Hajir–Maire tower-cutting argument", we likewise take those classical results as given.

6.2 This artifact does NOT construct the infinite tower

The existence of L_j with $[L_j : \mathbb{Q}] \to \infty$ and bounded root discriminant is supplied by Golod–Shafarevich [GS64] together with the Hajir–Maire splitting technique [HM01] adapted to pro-2 in [Alo+26]. This is a non-constructive existence theorem. No finite computation can exhibit the entire infinite tower. Phase 3 of the artifact stops at the boundary of computable admissibility.

6.3 This artifact has NOT been peer-reviewed

No external mathematician has reviewed this code base. Our confidence in the artifact rests on three independent signals:

1. the 59-test unit suite passes on CI (Ubuntu + Windows × Python 3.11 / 3.12);
2. the central numerical reproduction matches the published $\approx 6.24 \cdot 10^{-38}$ to 0.014% relative error (§5);
3. the source code is small ($\approx$ 1 000 LOC across four modules) and intended to be readable by anyone who has read [Alo+26] §2.

None of these substitutes for mathematician review. The artifact is offered as input to such review, not as a substitute.

6.4 Reported 0.014 and 0.0318 are NOT in either formal PDF

Secondary coverage (e.g. [Kal26]) reports numerical values $\delta \approx 0.014$ and $\delta \approx 0.0318$ in the informal contexts $n^{1.014}$ and $n^{1.0318}$. We verified by direct text extraction with `pypdf` that neither of these literal numerical values appears in either [Ope26] or [Alo+26]. The only rigorous numerical lower bound that appears verbatim in either formal PDF is $\delta_{\text{excess}} \approx 6.24 \cdot 10^{-38}$ from eq. (2.2) of [Alo+26].

The artifact's `SOURCE_CLAIMS` registry encodes this three-way provenance explicitly:

- `openai_index_reported` ($\delta = 0.014$, `verified_numerical_in_formal_pdf = False`),
- `sawin_rigorous_lower_bound` ($\delta = 6.24 \cdot 10^{-38}$, `verified_numerical_in_formal_pdf = True`),
- `sawin_refinement_reported` ($\delta = 0.0318$, `verified_numerical_in_formal_pdf = False`).

6.5 This artifact does NOT improve the published bound

We made no attempt to optimise the choice of T, S, L_T or the inner parameter k to obtain a larger δ_{excess}. The explicit choice of [Alo+26, §2.1] is what we encode and reproduce; the optimisation problem of finding (T, S, k) with larger admissible δ remains open and is out of scope.

7 Reproducibility

7.1 One-line reproduction

```
git clone https://github.com/Flamehaven-Labs/erdos-ant-verification
cd erdos-ant-verification
pip install -e ".[dev]"
python -m erdos_ant.verify
```

Expected terminal output:

```
Verdict:  PASS
Checks:  21/21 passed
eq (2.2) exponent excess:  6.2391e-38 (published ~6.24e-38, rel.err 0.0001)
(frozen-report mode:  tracked evidence not written; pass -write-evidence
to refresh)
```

By default the command does not write to disk; the working tree stays clean. Release maintainers pass `-write-evidence` to refresh the tracked files `reports/verification_result.json` and `reports/verification_report.md`.

7.2 Pinned environment

Item	Value
Repository	https://github.com/Flamehaven-Labs/erdos-ant-verification
Version (paper)	v0.1.3
Predecessor releases	v0.1.1 (source-only), v0.1.2 (pre-PDF)
Commit hash	<COMMIT_HASH> (filled in at v0.1.3 release-tag time)
Python	3.11 and 3.12 (CI-verified)
numpy	≥ 1.26
mpmath	≥ 1.3
Test count	60+ unit tests + 21 verify-script checks
CI matrix	Ubuntu $\times$ Windows $\times$ Python 3.11 / 3.12
Licence	MIT
DOI (predecessor, source-only)	10.5281/zenodo.20366884
DOI (this release)	<ZENODO_DOI> (filled in at v0.1.3 publish)

7.3 Independent third-party inspection

In addition to its own test suite, the artifact was scanned by three independent inspection tools, with the following results recorded at `reports/TRIPLE_INSPECTION_REPORT.md` in the repository:

Tool	Result	Headline metric
AI-SLOP-Detector v3.7.5	`clean` (6/6 files)	weighted deficit = 2.83 (threshold 30)
SPAR Framework v0.5.0	`ACCEPT` (Grade PASS)	score 100/100, claim drift = 0
SIDRCE SaaS v1.1.15	`CERTIFIED`	omega = 0.9289, trace chain verified

These cover structural-pattern review, claim-discipline review, and code-integrity certification respectively. None of them constitutes mathematical peer review; their relevance is bounded by the limits documented in §6.

7.4 Citation

A `CITATION.cff` file in the repository root provides citation metadata in the Citation File Format v1.2 standard, recognised by Zenodo, GitHub, and most citation managers. The recommended citation for a specific tagged release is the Zenodo DOI associated with that tag.

Acknowledgements

We thank the OpenAI internal reasoning model whose 2026 output is the subject of this artifact, and the authors of the remarks paper [Alo+26] for the simplified construction whose explicit numerical bound we reproduce: Noga Alon, Thomas F. Bloom, W. T. Gowers, Daniel Litt, Will Sawin, Arul Shankar, Jacob Tsimerman, Victor Wang, and Melanie Matchett Wood.

We thank Farshid Hajir, Christian Maire, and Ravi Ramakrishna for the class-field-tower-cutting technique referenced throughout the construction, and David A. Cox for the textbook treatment of genus theory that underlies the Phase 2 module.

This artifact was developed inside the Flamehaven-TOE TOE-TEST cell "Integrating Algebraic Number Theory Breakthroughs (Erdős–OpenAI)" and extracted to a standalone repository so that the verification surface could be independently installed, run, and cited without inheriting any Flamehaven-TOE governance layer. The Flamehaven-TOE Generative Discovery Charter informs the result-tagging discipline visible in the artifact's per-phase `result_tags` fields.

Inspection runs against the artifact's source were performed using the open-source Flamehaven audit stack: AI-SLOP-Detector, SPAR Framework, and SIDRCE SaaS. The findings are recorded in `reports/TRIPLE_INSPECTION_REPORT.md`.

References

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[Erd46] Paul Erdős. "On Sets of Distances of n Points". In: *American Mathematical Monthly* 53.5 (1946). Original statement of the unit-distance conjecture refuted in 2026., pp. 248–250.

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